

Final report

We began our project by writing out the guidelines for how we want our annotators to annotate the data. We had two different classification tasks assigned to us, one being whether something was technology, and the other being whether or not something was transportation related. Upon reviewing our annotations, we noticed quite a bit of disagreement within the technology data. We have hypothesized a couple of reasons for this. The first being that website links are very ambiguous as to whether or not they are technology related. We also think that some of the annotations could have been randomly input out of laziness of the annotator. A majority of the comments in the dataset were related to technology, since the comments were likely pulled using keywords like ‘Google’ or ‘Facebook’. Our second task, transportation, had a lot more agreement simply on the fact that the dataset was not targeted to be transportation related.

For our features, the first one we decided to use was fully capitalized words, meaning words that are entirely capitalized letters. We use this since fully capitalized words tend to either signify someone yelling or someone expressing a lot of emotion. Our second feature was the number of exclamation points in the comment. This is another good indicator of emotions like surprise, anger, and pain. For our final feature, we used the total number of repeated words in the comment. For example, if the comment was “San Antonio is really really big” then the repeated word count would be two.

Before starting our modeling, we needed to split the data using a train test split, with 80% training and 20% testing. We did this for both technology and transportation, giving us 8 variables for training and testing. For our model we used the LinearSVC model, which is a linear

classification model that was developed in scikit-learn. The first model that we ran was just with the comments being embedded into vectors using fit and fit transform.

		Technology	Transportation
Macro	Validation F1	0.5541	0.6089
	Precision	0.6116	0.7297
	Recall	0.6048	0.8477
	F1 score	0.6074	0.7710
Micro	Validation F1	0.7512	0.9012
	Precision	0.7300	0.8800
	Recall	0.7300	0.8800
	F1 score	0.7300	0.8800

We have some moderately good metrics for technology macro score (.607), and good metrics for the micro score (.730). Transportation overall has much better results, with a macro F1 score of 0.77, and a micro F1 score of 0.88.

Following this, we implemented the three features that we discussed earlier into the LinearSVC model. We implemented them individually first, one by one. Below are the results. The first set of columns looks at the capital words feature. The results are not so great for technology with macro and micro F1 scores of .29. Capital words seem to work a lot better for transportation, which has a macro F1 score of .78 and a micro F1 score of .88. Our second feature, the exclamation point, performed similarly to the capital letters feature. Our final feature, the number of repeated words in the comment, performed similarly for technology, but saw a slight performance drop for transportation.

Features		Capital words		Exclamation points		Repeat words	
		Technology	Transportation	Technology	Transportation	Technology	Transportation
Macro	Validation F1	0.6098	0.6098	0.6077	0.6077	0.6118	0.6118
	Precision	0.5016	0.7469	0.4982	0.7297	0.5041	0.6952
	Recall	0.5038	0.8561	0.4957	0.8477	0.5109	0.8288
	F1 score	0.2934	0.7869	0.2869	0.7710	0.2842	0.7371
Micro	Validation F1	0.9012	0.9012	0.9012	0.9012	0.9025	0.9025
	Precision	0.2850	0.8800	0.2900	0.8850	0.2750	0.8900
	Recall	0.2850	0.8800	0.2900	0.8850	0.2750	0.8900
	F1 score	0.2850	0.8800	0.2900	0.8850	0.2750	0.8900

		Technology	Transportation
Macro	Validation F1	0.5349	0.6093
	Precision	0.6058	0.7124
	Recall	0.6013	0.8386
	F1 score	0.6032	0.7544
Micro	Validation F1	0.7512	0.9025
	Precision	0.7350	0.8900
	Recall	0.7350	0.8900
	F1 score	0.7350	0.8900

Next, we decided to run the model with all three features being implemented, the results are above. We see great improvement to the model for technology when compared to running each of the features individually, with the F1 score improving to .60 for macro and .735 for micro. The transportation results remain approximately the same when compared to running the features individually. When comparing this table to our initial table, technology receives similar results to our initial model with just the reddit comments.

Random Forest		Technology	Transportation
Macro	Validation F1	0.4874	0.5359
	Precision	0.5614	0.5862
	Recall	0.8151	0.9385
	F1 score	0.5419	0.6143

Finally, we tried to implement another model to see how it would compare to the LinearSVC model. We decided to use the RandomForest model on our three features combined. The results are shown above. The results seem to be slightly worse than using the LinearSVC model.

To conclude, the transportation classification performs better than the technology classification. We think that the reason is because there are a lot more technology comments in the dataset when compared to transportation comments. Since there are many more comments flagged as technology comments than not, the model may be unable to pick up on what makes the comment technology related or not. Since transportation is only being flagged as yes in a few comments, we think it is much easier to predict. We think that if the sample had been larger, or if the sample had more diversity of different types of comments, the model may be able to pick up on that more easily.